move a marble (our solution) along the steepest path on the track (the landscape produced by the cost function), but stop in valleys. Figure 17.1 contrasts the gradient descent approach with the random search local optimization called *hill descent*[2]: in this case, the algorithm picks a direction randomly, and checks if by making a short (possibly random) step in that direction we get to an improvement: if so, it moves to the new solution; otherwise, no change is made, and another attempt is performed at the next iteration. As we have also explained in chapter 16, for gradient descent as well it is in theory possible to step over local minima, if the learning rate is large enough; this, however, is also unlikely to happen.